Protocol I8B-MC-ITSH

A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 and Humalog across Different Subcutaneous Doses in Healthy Subjects

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LY900014

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1. Protocol Synopsis

Title of Study:
A study to evaluate the pharmacokinetics and glucodynamics of LY900014 and Humalog across different subcutaneous doses in healthy subjects.

Rationale:
In clinical practice, patients with diabetes on insulin therapy will inject their prandial insulins at a range of individualised doses, which may or may not influence the absorption pharmacokinetics (PK), as well as, the glucodynamic (GD) effect. The primary objective of this study is to evaluate the PK and GD effects across 3 dose levels after administration of the commercial formulation of LY900014 in comparison to after administration of Humalog. It will also assess the dose proportionality of insulin lispro PK and GD after administration of the commercial formulation of LY900014.

Objectives/Endpoints:

Objectives	**Endpoints**
Primary To evaluate the insulin lispro PK following subcutaneous (SC) doses of LY900014 compared to Humalog (at doses of 7, 15, and 30 U) in healthy subjects.	Time to early half-maximal plasma concentration (early 50% t_{max}) and area under the concentration versus time curve (AUC) from time zero to 30 minutes
Secondary 1. To evaluate the dose proportionality of the 3 dose levels of LY900014 as assessed by insulin PK and GD measures.	1a. PK: AUC from time zero to 10 hours (AUC[0-10h]), AUC from time zero to infinity (AUC[0-∞]), and maximum observed drug concentration (C_{max}) 1b. GD: Total amount of glucose infused (G_{tot}) and maximum glucose infusion rate (R_{max})
2. To evaluate the GD of LY900014 compared to Humalog following SC doses of 7, 15, and 30 U in healthy subjects using a euglycaemic clamp procedure.	2. Time to half-maximal GIR before tR_{max} (early 50% tR_{max}), G_{tot} over 30 minutes (G_{tot}[0-30min]), G_{tot} over 1 hour (G_{tot}([0-1h]), and time to onset of insulin action (T_{onset})
3. To evaluate the tolerability of LY900014 in healthy subjects.	3. Adverse events (AEs) and injection-site reactions

Summary of Study Design:

Study I8B-MC-ITSH is a Phase 1, randomised, subject- and investigator-blind, 6-period complete crossover study in up to 42 healthy subjects to evaluate the PK and GD characteristics of LY900014 doses in comparison to Humalog at doses of 7, 15, and 30 U.

Treatment Arms and Planned Duration for an Individual Subject:
Subjects will be randomised to 1 of 6 treatment sequences to assess different doses of LY900014 (7, 15, and 30 U) and Humalog (7, 15, and 30 U). The study will include a 28-day screening period, followed by 6 study periods, with

a wash-out period of at least 3 days between each study period. The follow-up visit will take place at least 14 days after the last dose.

Number of Subjects:
Up to 42 healthy men and women may be enrolled to target approximately 34 subjects to complete.

Statistical Analysis:

The primary statistical analyses for PK will be conducted on those subjects who receive at least 1 dose of study drug and have measurable insulin lispro concentrations. The primary statistical analysis for GD will be conducted on those subjects who complete at least 1 clamp procedure. Supportive analyses will be done on the key parameters for the subjects who complete all treatment periods with evaluable data.

Safety:

Safety analyses will be conducted for all enrolled subjects, whether or not they complete all protocol requirements. All investigational product and protocol-procedure AEs will be listed, and if the frequency of events allows, safety data will be summarised using descriptive methodology. Safety parameters that will be assessed include safety laboratory parameters and vital signs, as well as evaluation of injection-site reactions. The parameters will be listed and may be summarised using standard descriptive statistics.

Pharmacokinetics:

Pharmacokinetic analyses will be conducted using standard noncompartmental methods of analysis. Pharmacokinetic parameters will be assessed using free serum insulin lispro.

Log-transformed AUC estimates and C_{max} for insulin lispro will be evaluated to estimate least-squares geometric means, ratios of geometric means between LY900014 and Humalog for each dose level, and their corresponding 95% CIs using the mixed-effects model that includes treatment, dose level, dose level-by-treatment interaction, and period as fixed effects and subject as a random effect.

The same model without log transformation will be used for the analysis of the PK time parameters (time to early half-maximal plasma concentration [early 50% t_{max}], time to late half-maximal plasma concentration [late 50% tmax], time to maximum observed drug concentration [t_{max}], and half-life associated with the terminal rate constant in noncompartmental analysis [$t_{1/2}$]). Least-squares means (LSmeans), treatment differences in LSmeans, and the corresponding 95% confidence intervals (CIs) for the treatment differences for each dose level will be estimated from the model. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

The analyses described above will also be performed on the population of subjects who completed and had evaluable PK data for all study periods

The degree of dose proportionality for the insulin lispro in LY900014 will be assessed by fitting the power model to both AUC (AUC[0-10h], and AUC[0-∞]) and C_{max} versus dose for each dose level of LY900014. The estimated ratio of dose-normalised geometric means of PK parameters between the highest and lowest doses will be used to assess dose proportionality. In addition, the slope and its 95% CIs and the geometric LSmeans for each dose level tested will be produced. In the event that the power model is not a good representation of the data over the entire dose range tested, alternative models may be investigated. Log-transformed C_{max} and AUC LSmeans, and 95% CI estimates for each dose will be back-transformed to provide the geometric means and the corresponding 95% CIs.

The analyses will also be performed using the subset of the subjects who complete all treatment periods with evaluable data.

Glucodynamics:

Glucodynamic assessments will be determined from the glucose clamp procedure, where the glucose infusion rate (GIR) over time will be used as a measure of insulin effect. Glucodynamic analyses will be conducted on those patients who complete at least 1 clamp procedure. A locally weighted scatterplot smoothing function will be applied to all individual GIR versus time profiles in each treatment group and/or period. The fitted data for each patient will be used to calculate the GD parameters.

The GD statistical model will be the same as the model used for the analysis of the PK parameters. The following variables will be log transformed prior to analysis: R_{max}, G_{tot}, G_{tot}(0-30min), G_{tot}(0-1h), and G_{tot} over 10 hours.

The same model without log transformation will be used for the analysis of the GD time parameters (T_{onset}, time to maximum GIR [tR_{max}], early 50% tR_{max}, time to half-maximal GIR after tR_{max}). Least-squares means, treatment differences in LSmeans, and the corresponding 95% CIs for the treatment differences for each dose level will be estimated from the model. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

The degree of dose proportionality for the GD of LY900014 will be assessed by fitting the power model to G_{tot} and R_{max} parameters. The estimated ratio of dose-normalised geometric means of PK parameters between the highest and lowest doses will be used to assess dose proportionality. In addition, the slope and its 95% CIs and the geometric LSmeans for each dose level tested will be produced. In the event that the power model is not a good representation of the data over the entire dose range tested, alternative models may be investigated. Log-transformed G_{tot} and R_{max} LSmeans, and 95% CI estimates for each dose will be back transformed to provide the geometric means and the corresponding 95% CIs. The degree of dose proportionality in GD following administration of LY900014 will be assessed by fitting the power model to G_{tot} and R_{max} versus dose for each dose level of LY900014.

Pharmacokinetics and GD analyses will also be performed using the subset of the subjects who complete all treatment periods with evaluable data.

2. Schedule of Activities

Study Schedule for Protocol I8B-MC-ITSH

Procedures	Screening	Euglycaemic Clamp Procedure Periods 1 to 6		FU/ ED[a]	Comments
	Up to Day -28	Day -1	Day 1		
Informed consent	X				Must occur before screening procedures. Screening procedures should take place no later than 28 days after signing the informed consent.
Admission to CRU		X			Enrolment will occur on the first admission.
Overnight fast	X	X			Subjects are expected to fast for approximately 8 hours before screening, and at least 8 hours starting from the evening before each dose until the end of the glucose clamp procedure.
Weight	X	X			Day -1 weight for Period 1 only.
Height	X				
Hip and waist circumference		X			Period 1 only. Record the average of triplicate measurements.
Physical examination/ Medical assessment[b]	X		Predose and before discharge from CRU for each period	X	Full physical examination and medical history at screening. Thereafter, medical review and targeted examination, as appropriate and as deemed necessary by the investigator.
Vital signs (supine): blood pressure and pulse rate[b]	X		Predose and after the end of clamp procedure	X	
12-Lead ECG[b]	X		Predose and after the end of clamp procedure	X	Single ECGs to be collected for safety.
Clinical laboratory tests[b]	X		Predose of Period 1 only	X	Results not required prior to first dosing.
Pregnancy test/urine drug screen/alcohol breath test	X	X		X	Pregnancy test for female subjects of childbearing potential only. Serum pregnancy test at screening. Urine pregnancy test for all other visits. Alcohol breath test and drug screen may be omitted at follow up.
Study drug administration			0 minute		Time of study drug administration = 0 minute. Study drug will be administered at about the same time on Day 1 of each study period.
Injection-site assessments			0, 60, 240, and 600 minutes		Assessment times are relative to study drug administration time (time 0). The 0-minute assessment will start as soon as practicably possible (within 1 minute) following the injection.
Insulin lispro PK sampling			0, 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 90, 120, 150, 180, 210, 240, 300, 360, 420, 480, 540, and 600 minutes		Sampling times are relative to study drug administration time (time 0). The 0-minute sample should be taken within 2 minutes prior to planned dosing.

Procedures	Screening	Euglycaemic Clamp Procedure Periods 1 to 6		FU/ ED[a]	Comments
	Up to Day -28	Day -1	Day 1		
Euglycaemic clamp and blood glucose sampling for procedure			The clamp device will automatically sample blood at a rate of 2 mL/h after connection and record a blood glucose value every minute during the clamp procedure. A manual blood sample will also be taken at least every 30 minutes during the clamp for checking the correct functioning of the clamp device blood glucose readings and for safety.		Blood glucose monitoring will be done every minute starting from calibration after connection to the clamp device, baseline glucose establishment, drug dosing, and clamp up to 10 hours after dosing
C-peptide sampling			0, 60, 120, 240, 360, 480, and 600 minutes		Sampling times are relative to study drug administration (time 0). The 0-minute sample should be taken within 2 minutes prior to planned dosing.
Pharmacogenetic sample		X			Period 1 only.
Immunogenicity sample[b]			Predose for Period 1, Period 3, and Period 5	X	Additional samples may be collected if the investigator considers there is a possibility that an adverse event is immunologically indicated.
Discharge from CRU			X		After completion of all study procedures and medical assessment on Day 1. Investigator's discretion to discharge on Day 2.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; ED = early discontinuation; FU = follow-up; PK = pharmacokinetics.

a Procedures should be performed at least 14 days after the last dose of study drug.

b Predose assessments, procedures, and sampling may be performed any time before actual dosing on Day 1.

Note: The site should schedule activities as appropriate. In cases when several study procedures are scheduled at the same time, the order of priority will be as follows: PK samples, glucose samples, injection-site assessments, vital signs, ECG, clinical laboratory samples, immunogenicity, C-peptide, and pharmacogenetic samples. If the investigator decides based on clinical judgement not to dose a subject on a given day (for example, because of low blood glucose), the subject's visit may be rescheduled; any procedures performed in that period may be repeated.

3. Introduction

3.1. Study Rationale

In clinical practice, patients with diabetes on insulin therapy will inject their prandial insulins at a range of individualised doses, which may or may not influence the absorption pharmacokinetics (PK), as well as, the glucodynamic (GD) effect. The primary objective of this study is to evaluate the PK and GD effects across 3 dose levels after administration of the commercial formulation of LY900014 in comparison to Humalog®. It will also assess the dose proportionality of insulin lispro PK and GD after administration of the commercial formulation of LY900014.

3.2. Background

The insulin analogue, insulin lispro (Humalog; Eli Lilly and Company), has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2015). In healthy volunteers given subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 U/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). However, the general consensus is that rapid-acting insulin, administered either through pumps or syringes/pen injectors, is still not rapid enough to match carbohydrate absorption profiles, which limits efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the PK and GD profiles so that they have an even faster onset to better match carbohydrate absorption and also may allow important flexibility in the time of dosing relative to meals.

LY900014 is an ultra-rapid-acting insulin lispro formulation with treprostinil CCI [redacted] and other ingredients. LY900014 has an increased early absorption of insulin lispro compared to commercially available insulin lispro (Humalog). CCI [redacted]

Treprostinil is a prostacyclin analogue, administered either through inhalation CCI [redacted], as an intravenous (IV) infusion, or as a continuous SC administration for the treatment of symptomatic pulmonary arterial hypertension (PAH) and has been approved in the US since 2002 CCI [redacted]. Each of the other excipients (such as sodium citrate and magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration (FDA)'s Generally Recognized as Safe Food Additives database and in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Safety and tolerability of LY900014 have been demonstrated in healthy subjects in 3 previous clinical studies, in approximately 71 healthy subjects across a dose range of 7.5 to 30 U. All 3 studies were Phase 1, randomised, subject-blind studies in which the PK and GD of LY900014 and Humalog were evaluated during a euglycaemic glucose clamp following SC administration. The total insulin lispro exposure and GD effect were similar for LY900014 and Humalog insulin

lispro; however, LY900014 demonstrated a faster and earlier insulin lispro absorption compared to Humalog. LY900014 also displayed a dose proportional increase in the insulin lispro exposure (maximum observed drug concentration [C_{max}] and area under the concentration versus time curve from time [AUC]) across this dose range.

In addition, data from two Phase 1b studies showed LY900014 was well tolerated in patients with type 1 diabetes mellitus (T1DM; 30 patients) and type 2 diabetes mellitus (T2DM; 30 patients) using multiple daily injections (MDIs). There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events (TEAEs) were reported, and there were no notable increases in these events in relation to any of the LY900014 formulations compared to those in relation to Humalog.

More information can be found in the Investigator's Brochure (IB) of LY900014.

3.3. Benefit–Risk Assessment

Study I8B-MC-ITSH (ITSH) will not offer any direct benefits to the healthy subjects participating in the study. The data from previous studies in healthy subjects and data from studies in patients with T1DM and T2DM have shown that LY900014 was well tolerated and the adverse drug reactions are in keeping with those reported for Humalog.

Potential risks associated with LY900014, derived from the known risks of insulin lispro (Humalog), are hypoglycaemia, hypersensitivity reactions (localised allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral oedema (Humalog package insert, 2015).

Notably, across all doses in the Lilly clinical studies with healthy subjects and patients with diabetes that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, CCI Moreover, data from Lilly Phase 1b patient studies showed that only 1 out of 472 plasma samples had detectable treprostinil levels taken from a total of 30 patients with T1DM (doses up to CCI per SC bolus injection) and 30 patients with T2DM (doses up to CCI per SC bolus injection) following MDI administration of LY900014. The exposure levels of treprostinil from the doses of LY900014 planned in this study are thus expected to be undetectable and substantially lower than those observed in the treatment of PAH. In preclinical safety pharmacology and toxicity studies or clinical pharmacology studies involving LY900014 or treprostinil alone, other than known risks associated with Humalog and CCI , no additional risks were identified. No known potential risks are associated with the use of small amounts of treprostinil CCI in the LY900014 formulation. Additionally, local and systemic toxicity profiles of Humalog and CCI do not suggest the potential for additive or synergistic toxicity.

Following administration of the study insulin, subjects will receive IV glucose infusion at a variable rate to maintain euglycaemia up to 10 hours after insulin lispro administration. The aim of the clamp procedure is to maintain blood glucose within the normal glycaemic range. In

addition, the clamp is performed while the subject is inpatient and under the investigator's supervision. These considerations should minimise the risk of hypoglycaemia in subjects participating in Study ITSH.

More information about the known and expected benefits, risks, SAEs, and reasonably anticipated AEs of LY900014 is to be found in the IB.

4. Objectives and Endpoints

Table ITSH.1 shows the objectives and endpoints of the study.

Table ITSH.1. Objectives and Endpoints

Objectives	Endpoints
Primary To evaluate the insulin lispro pharmacokinetics following SC doses of LY900014 compared to Humalog (at doses of 7, 15, and 30 U) in healthy subjects.	Early 50% t_{max} and AUC(0-30min)
Secondary 1. To evaluate the dose proportionality of the 3 dose levels of LY900014 as assessed by insulin lispro pharmacokinetics and glucodynamic measures.	1a. PK: AUC(0-10h), AUC(0-∞), and C_{max} 1b. GD: G_{tot} and R_{max}
2. To evaluate the GD during euglycaemic clamp procedure following SC doses of 7, 15, and 30 U of LY900014 compared to Humalog in healthy subjects.	2. Early 50% tR_{max}, G_{tot}(0-30min), G_{tot}(0-1h), and T_{onset}
3. To evaluate the tolerability of LY900014 in healthy subjects.	3. AEs and injection-site reactions
Exploratory 1. Explore the formation of antibodies to insulin lispro.	1. Anti-insulin lispro antibodies
2. To assess C-peptide levels following administration of LY900014 and Humalog.	2. C-peptide concentration

Abbreviations: AE = adverse event; AUC(0-30min) = area under the concentration versus time curve from time zero to 30 minutes; AUC(0-10h) = AUC from time zero to 10 hours; AUC(0-∞) = AUC from time zero to infinity; C_{max} = maximum observed drug concentration; early 50% t_{max} = time to early half-maximal plasma concentration; early 50% tR_{max} = time to half-maximal glucose infusion rate before tR_{max}; GD = glucodynamics; G_{tot} = total amount of glucose infused; G_{tot}(0-30min) = G_{tot} over 30 minutes; G_{tot}(0-1h) = G_{tot} over 1 hour; PK = pharmacokinetics; R_{max} = maximum glucose infusion rate; SC = subcutaneous; T_{onset} = time to onset of insulin action.

5. Study Design

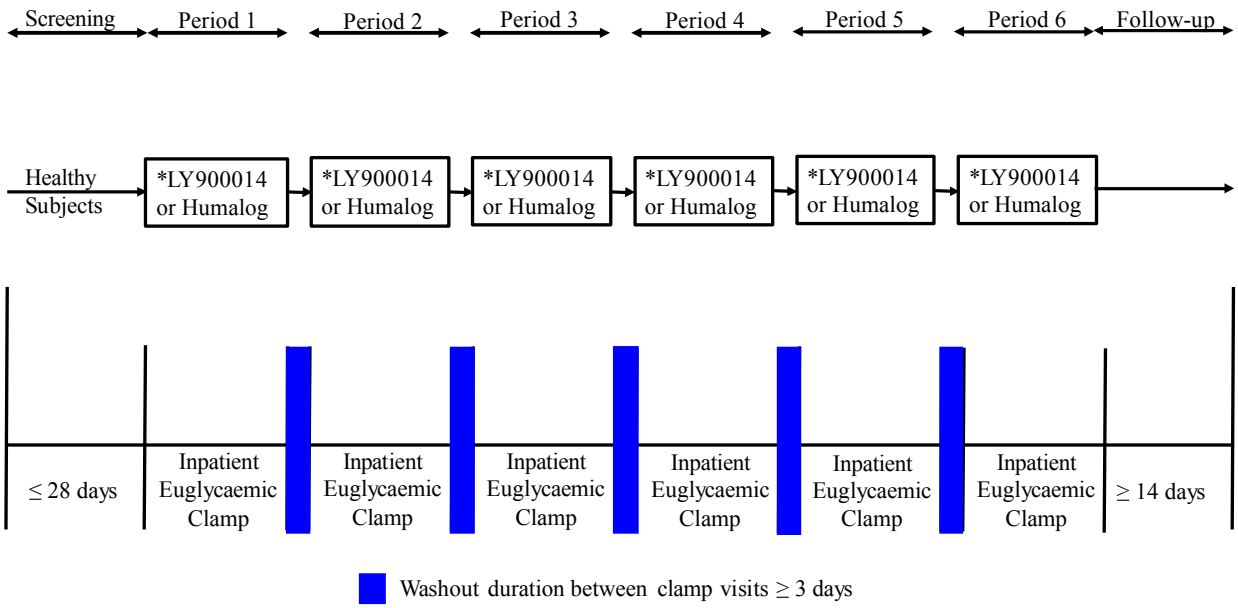
5.1. Overall Design

This is a Phase 1, randomised, subject- and investigator-blind, 6-period complete crossover study in up to 42 healthy subjects to evaluate the PK and GD characteristics across a range of LY900014 doses.

Approximately 34 subjects are expected to complete the study. Each subject will be randomised to 1 of 6 treatment sequences comprising single SC doses of 7, 15, and 30 U of LY900014 or Humalog administered SC in the abdominal wall (Table ITSH.2). A minimum of 3 days between study dosing of each consecutive study period will be required for an individual subject.

Figure ITSH.1 illustrates the study design.

Study governance considerations are described in detail in Appendix 3.



*Single dose of 7, 15, or 30 U of LY900014 or Humalog administered subcutaneously to the abdomen. Euglycaemic clamp procedure after each dose.

Figure ITSH.1. Illustration of study design for I8B-MC-ITSH.

Table ITSH.2. Treatment Sequence Example for I8B-MC-ITSH

Treatment Sequence	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
1	LY 7 U	LY 15 U	H 30 U	LY 30 U	H 15 U	H 7 U
2	LY 15 U	LY 30 U	LY 7 U	H 7 U	H 30 U	H 15 U
3	LY 30 U	H 7 U	LY 15 U	H 15 U	LY 7 U	H 30 U
4	H 7 U	H 15 U	LY 30 U	H 30 U	LY 15 U	LY 7 U
5	H 15 U	H 30 U	H 7 U	LY 7 U	LY 30 U	LY 15 U
6	H 30 U	LY 7 U	H 15 U	LY 15 U	H 7 U	LY 30 U

Abbreviations: H = Humalog; LY = LY900014.

Note: This is only an example table for illustration purpose; subjects will be assigned a treatment sequence according to the actual treatment randomisation schedule provided to the unblinded site pharmacist.

Subjects will be admitted to the clinical research unit (CRU) on the evening before each dosing day and will remain in the CRU for the duration of the clamp period and until discharge by the investigator. Subjects are expected to fast for at least 8 hours before each dose. Following dose administration, each subject will undergo an euglycaemic clamp procedure of up to 10 hours. Upon completion of the clamp procedures, the subjects will be provided a meal and discharged from the CRU in the evening after medical assessments. Subjects may remain in the CRU if deemed necessary for safety monitoring, as determined by the investigator.

Subjects will be required to attend the CRU on at least 8 occasions:

- screening visit (up to 28 days prior to dosing)
- six treatment visits for the dosing and clamp procedure (study Periods 1 to 6) with a minimum period of at least 3 days between consecutive clamp visits
- follow-up visit (at least 14 days after the last dose), or early discontinuation

In the study, safety assessments and an evaluation of local tolerability at injection sites will be performed as specified in the Schedule of Activities (Section 2). Local tolerability assessments will include injection-site reactions within the following categories: pain on palpation, itching, erythema, oedema, and induration/infiltration.

5.2. Number of Participants

A total of 42 subjects may be enrolled for an approximately 34 completing subjects. For purposes of this study, a subject is considered as having completed the study when all scheduled procedures shown in the Schedule of Activities (Section 2) have been completed. If a subject discontinues before completion of all 6 dosing periods, replacement subjects may be enrolled. The replacement subject will be assigned the treatment sequence of the discontinued subject and complete that sequence in its entirety.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last subject.

5.4. Scientific Rationale for Study Design

A population of healthy subjects is selected based on the likelihood of less physiologic variability in the absence of disease states that may affect multiple organ systems and absence of other confounding factors such as concomitant medications.

The use of a crossover design allows each subject to serve as his or her own control, thereby reducing variability. The study is subject- and investigator-blinded to minimise potential bias related to the clamp procedure.

A euglycaemic clamp technique, the gold standard methodology for assessing insulin action, will be used in this study to provide data on the GD activity of LY900014 (see Section 9.6 for a detailed description of the clamp methodology) relative to Humalog. Previous studies have shown that the 10-hour duration of the euglycaemic glucose clamp will allow for a complete assessment of GD activity of LY900014.

Based on the PK properties of treprostinil (half-life associated with the terminal rate constant in noncompartmental analysis [$t_{1/2}$] = CCI and Humalog ($t_{1/2}$ = CCI, a minimum duration of 3 days for the washout period between clamp visits and at least 14-days duration between the last dose of study drug and the follow-up visit are considered appropriate.

In patients with diabetes, mealtime insulin requirements can vary substantially depending on carbohydrate content, glycaemic index of the meal, and the insulin sensitivity of patient. Understanding the PK and GD across a broader dose range is important as reliable titration of an insulin product depends on an established dose–response relationship.

5.5. Justification for Dose

The safety, PK, and pharmacology of LY900014 at similar doses and with the similar formulation composition have been assessed in clinical studies in healthy subjects, in patients with T1DM and T2DM using MDI. In addition, clinical studies had evaluated the safety, PK, and pharmacology of SC bolus doses of treprostinil alone. More information can be found in the CCI.

All tested doses of LY900014 (up to 50 U) and treprostinil (CCI) were well tolerated in healthy subjects, patients with T1DM, and patients with T2DM. There were no SAEs related to study treatment in any of the 8 studies. No subject discontinued from the studies because of drug-related AEs. Trial participants in these studies were monitored for changes in vital signs; there were no significant systemic haemodynamic effects of treprostinil at the doses administered based on blood pressure and heart rate. Visual analogue scale pain scores showed that the SC injections of insulin lispro plus treprostinil co-formulations and treprostinil alone were well tolerated. In summary, there were small numbers of TEAEs and injection-site AEs, but there was no clinically significant increase in these or other events compared to placebo or to Humalog and no clinically significant increase in frequency with higher doses of treprostinil. Notably, at the higher doses of treprostinil, there were no clinically significant increase in those AEs associated with systemic absorption as described in the CCI (i.e.,

headache, diarrhoea, nausea, jaw pain, vasodilatation, rash, oedema, and hypotension). This is likely attributable to the relatively lower exposures of treprostinil demonstrated in these clinical studies.

Study ITSH will evaluate 3 insulin lispro dose levels (7, 15, and 30 U), which are within the range of insulin doses that are used for the treatment of both patients with T1DM and T2DM in clinical practice. At the highest dose level of 30 U of LY900014, subjects will be expected to receive approximately CCI treprostinil in a single SC dose, which is within the range evaluated as safe and tolerated in previous studies. The PK profile of treprostinil showed that concentrations of treprostinil in healthy subjects (after administration of up to 30 U of an experimental LY900014 formulation which contained CCI of treprostinil) were quantifiable in only CCI of the samples at 15 minutes postdose and CCI of the samples at 30 minutes postdose occurring around C_{max}. Among subjects with detectable levels, the mean concentrations were only slightly above the detection limit of the assay (lower limit of quantitation = CCI), and there were no detectable levels in any samples at 1 and 2 hours after injection. Thus, the expected treprostinil C_{max} following LY900014 administration would be CCI than the steady-state treprostinil levels in patients with PAH treated with CCI CCI from an average dose of CCI of CCI in clinical trials CCI , CCI .

6. Study Population

Eligibility of subjects for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any pre-existing conditions will be documented.

Screening may occur up to 28 days prior to enrolment. Subjects who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

6.1. Inclusion Criteria

Subjects are eligible for inclusion in the study only if they meet all of the following criteria at screening:

[1] are overtly healthy males or females, as determined by medical history and physical examination

[1a] male subjects: agree to use an effective method of contraception for the duration of the study and for 1 month following the last dose of investigational product (IP)

[1b] female subjects:

- women of childbearing potential may participate and include those who test negative for pregnancy before initiation of treatment based on a urine pregnancy test and agree to use 1 highly effective method of contraception or a combination of 2 effective methods of contraception during the study and for 1 month following the last dose of the IP
- women of nonchildbearing potential may participate in the study without using adequate contraceptive methods, and include those who are:
  - infertile due to surgical sterilisation (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as müllerian agenesis; or
  - postmenopausal, defined as women aged <52 years and being amenorrhoeic for more than 1 year with a serum follicle-stimulating hormone (FSH) level compatible with postmenopausal status (positive FSH level) or aged ≥52 years and being amenorrhoeic for less than 1 year and with a serum FSH level compatible with postmenopausal status or aged ≥52 years being amenorrhoeic for more than year.

[2] are between 18 and 65 years of age, inclusive, at the time of screening.

[3] have a body mass index (BMI) of 18.0 to 30.0 kg/m^2, inclusive, at the time of screening.

[4] have clinical laboratory test results within normal reference range for the population or CRU, or results with acceptable deviations that are judged to be not clinically significant by the investigator.

[5] have venous access sufficient to allow for blood sampling, IV administration, and clamp procedure as per the protocol.

[6] are nonsmokers, have not smoked for at least 6 months before entering the study, and agree not to smoke (cigars, cigarettes, or pipes) or use smokeless tobacco or nicotine products for the duration of the study.

[7] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures.

[8] are able and willing to give signed informed consent.

6.2. Exclusion Criteria

Subjects will be excluded from study enrolment if they meet any of the following criteria at screening and/or enrolment:

[9] are CRU personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling.

[10] are Lilly employees or employees of the CRU.

[11] are currently enrolled in a clinical study involving an IP or any other type of medical research judged not to be scientifically or medically compatible with this study.

[12] have participated, within the past 30 days, in a clinical study involving an IP. If the previous IP has a long half-life, 5 half-lives, or 3 months (whichever is longer) should have passed.

[13] have previously completed or withdrawn from this study.

[14] have known allergies to treprostinil or insulin lispro, related compounds, or any components of the formulation, or a history of significant atopy.

[15] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study.

[16] have an abnormal blood pressure and/or pulse rate as deemed to be clinically significant by the investigator.

[17] have a history or presence of cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine, haematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the study drug; or of interfering with the interpretation of data.

[18] have known or ongoing psychiatric disorders deemed to be clinically significant by the investigator.

[19] regularly use known drugs of abuse and/or show positive findings on urinary drug screening.

[20] show evidence of human immunodeficiency virus (HIV) infection, and/or positive human HIV antibodies.

[21] show evidence of hepatitis B and/or positive hepatitis B surface antigen.

[22] show evidence of hepatitis C and/or positive hepatitis C antibody (the presence of hepatitis C antibodies in the setting of normal liver function tests and a negative hepatitis C polymerase chain reaction are not an exclusion).

[23] use over-the-counter or prescription medication within 7 to 14 days, respectively, prior to dosing apart from vitamin/mineral supplements, occasional paracetamol, topical treatments, thyroid replacement medication, or birth control methods and throughout the study. If this situation arises, inclusion of an otherwise suitable subject may be at the discretion of the investigator and sponsor.

[24] have donated blood or have blood loss of more than 500 mL within the previous 3 months of study screening.

[25] have a significant history of alcoholism or drug abuse as judged by the investigator or consuming more than 24 g of alcohol per day for males or more than 12 g of alcohol per day for females or are unwilling to stop alcohol consumption at least 24 hours before each CRU admission (Day -1) and throughout the duration of each CRU visit.

[26] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, subjects may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

For all treatment periods, subjects will fast for at least 8 hours prior to dosing and until the glucose clamp procedure is completed, after which subjects will receive a meal. Water can be consumed freely during this time.

When not a resident at the CRU, subjects will be encouraged to follow their normal diets.

6.3.2. Caffeine, Alcohol, and Tobacco

No alcohol intake will be allowed at least 24 hours before each CRU admission and while resident within the CRU.

Subjects should refrain from caffeine-containing food/beverages (for example, cola, chocolate, tea, and coffee) for at least 12 hours before each dose and throughout the duration of each CRU visit.

Smoking (cigars, cigarettes, or pipes) and the use of smokeless tobacco products will not be permitted during the study.

6.3.3. Activity

Subjects are encouraged to maintain their regular exercise habits; however, they should not undertake vigorous or prolonged exercise within 48 hours prior to dosing. These subjects will have their dosing visits deferred or may be excluded from this study, as judged by the investigator to prevent interference with study results. After dosing, subjects should remain recumbent or sitting in the CRU until the end of the glucose clamp.

Movement will be restricted to retain the integrity of connections to the infusion(s) and the study procedures.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be re-screened.